AI ACADEMY · LEVEL 4 · GRADE 9 · WEEK 32 · STUDENT LESSON

Proxy Features and Fairness

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Responsible Plant Classifier

Before you start

Week 30: a shortcut can correlate with labels without representing the intended concept.

Plan time to read, investigate and explain. Keep predictions separate from observations.

STUDENT LESSON · 1 OF 18

Your mission

> Start with the target: Identify possible proxy features. Try the recall before opening notes.

Your learning target

I can identify possible **proxy feature**s.

I can explain why removing one field does not guarantee **fairness**.

I can propose a contextual review with concrete **evidence**.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Subgroup Performance. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Feature Engineering. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Reproducible Experiments. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

A feature may indirectly encode sensitive identity or context, creating unjustified differences even when the original field is absent.



Your target

LEARNING TARGET A leaf model performs well using location ID alone. Explain a possible proxy problem and design a diagnostic evaluation.

Builder name and experiment date:

Readiness for the connected explanation

Does equal overall accuracy prove fair treatment?

This week's end goal

By the end, I can explain **proxy feature**s and **fairness** using a traced example and a stated limitation.

STUDENT LESSON · 2 OF 18

Notice the evidence

> Investigate the supplied Proxy Features and Fairness case. Separate observations, given facts and assumptions before choosing an explanation.

PICTURE ENTRY

Notice → Predict → Explain

Begin with visible **evidence**; do not explain more than the picture and information support.



Start with this puzzle

MYSTERY The team must use **proxy feature**s and **fairness** to solve a model-building problem. What information do they have? What important **evidence** is still missing?

Name three visible details without interpreting them.

Predict the data, model, or evaluation decision being made.

Identify one hidden assumption that could change the result.

State what **evidence** would confirm or reject your **prediction**.

My **prediction** and supporting **evidence**:

A closer explanation

An **input** can carry information about a characteristic without naming it. In a fictional school-device dataset, device code may be associated with building or access to newer equipment.

Removing a building field does not remove all related information. Do not infer a real student identity or protected characteristic in this exercise.

Use fictional classroom data throughout this investigation. Keep the task, units, reference labels and data split visible as you reason.

Earlier weeks supplied the model-building **process**; this week examines **proxy feature**s and **fairness**.

STUDENT LESSON · 3 OF 18

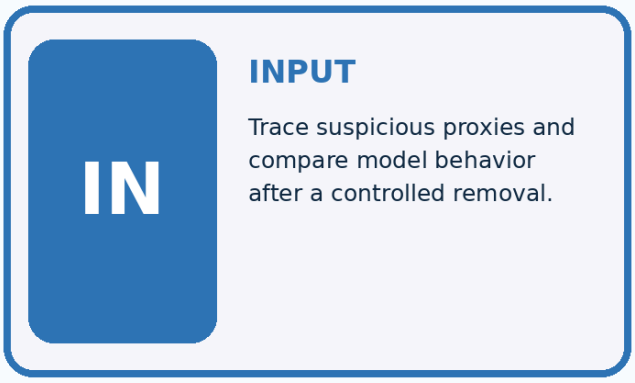
See how it works

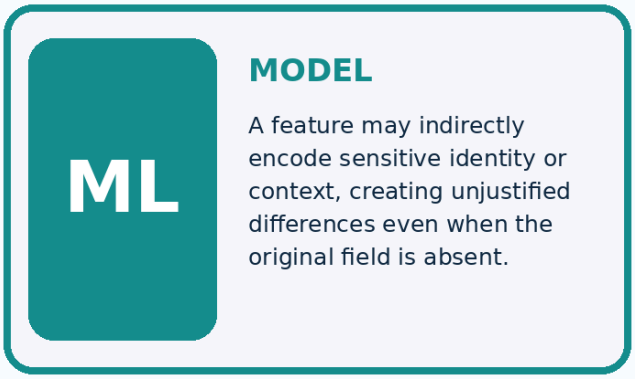
> Follow the mechanism for Proxy Features and Fairness. Connect each labelled input, deciding step and result.

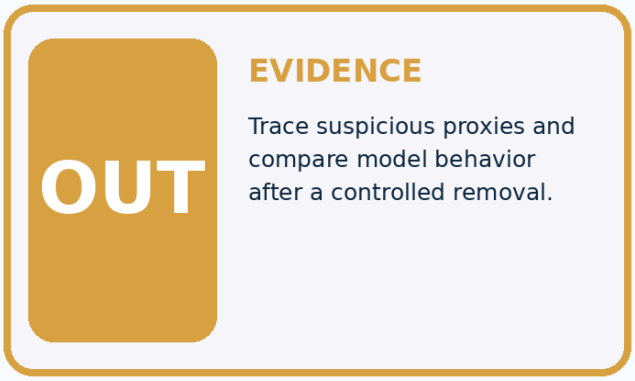
ML MECHANISM

Input → Model → Evidence

Trace the information, fitted behavior, and evaluation proof separately.







Question to explain

QUESTION Trace suspicious proxies and compare model behavior after a controlled

removal.

MODEL IDEA A feature may indirectly encode sensitive identity or context, creating unjustified differences even when the original field is absent.

PROOF NEEDED Use an unseen case, visible calculation, error analysis, and a stated limitation.

How the mechanism works

UNDER THE HOOD Photo background predicts classroom location and is correlated with camera quality. Model uses background as a shortcut for health labels.

Draw a fresh data → model → **prediction** → evaluation trace:

A closer explanation

Fictional records show device code X on 8 of 10 north-building examples and 2 of 10 south-building examples. Device and building are associated in this sample.

This does not prove device causes model errors or that every X device comes from north. The remaining cases contradict such a universal rule.

Before continuing, underline the decision rule and circle the **evidence** that supports the conclusion. Explain the example to a partner without replacing the counts or conditions with a stronger claim.

Keep the distinction between an observed result and a proposed next step.

Connect the picture and the explanation

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

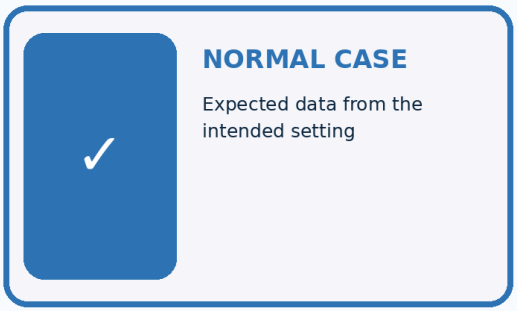
Compare the cases

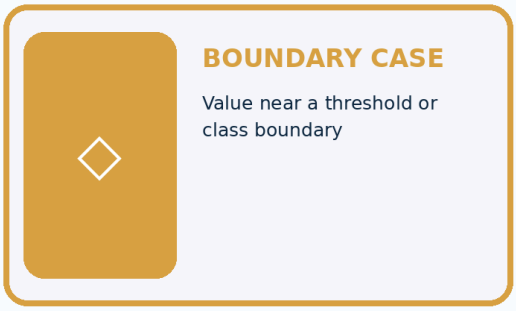
> Compare the supplied cases for Proxy Features and Fairness. Identify the detail that changes the conclusion.

EVALUATION DESIGN

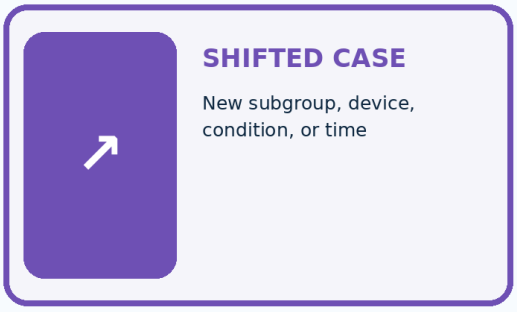
Four Tests, Four Kinds of Evidence

A system is not understood until it survives more than one comfortable example.









NORMAL Expected data and expected behavior.

Keep this boundary clear

BOUNDARY Value close to a threshold, ambiguous label, or uncertain **prediction**.

MISSING Necessary field, label, source, or measurement is unavailable.

SHIFTED New lighting, species, device, subgroup, or time condition.

RELEASE RULE Passing one case never proves generalization. Predefine expected behavior for all four categories.

Which case most threatens today's claim, and why?

Change one thing in the pictured case

Change one thing

Change: Camera codes are assigned independently of setting in a new balanced dataset.

Predicted result

Prediction: camera_code no longer perfectly identifies setting in that dataset.

Why the result follows

Reason: The observed mapping changed; other retained fields may still contain setting information.

STUDENT LESSON · 5 OF 18

Words that unlock the idea

> Use the vocabulary of Proxy Features and Fairness to explain a concrete decision, rather than only memorising definitions.

VOCABULARY

Words You Can Calculate and Defend

A definition matters when it lets you interpret a new dataset, model, or result.

1

PROXY FEATURE

input correlated with another sensitive attribute

2

FAIRNESS

evaluate benefits and harms across affected groups

3

PAIRED TEST

cases differing in one relevant factor

4

MITIGATION

control reducing an identified risk

Term	Precise meaning	My ML example
proxy feature	input correlated with another sensitive attribute	
fairness	evaluate benefits and harms across affected groups	
paired test	cases differing in one relevant factor	
mitigation	control reducing an identified risk	

MISCONCEPTION TO REPAIR Removing a sensitive column guarantees the model cannot use related information.

Repair the claim using a definition, calculation, or test record:

Words used in the closer explanation

Word	Meaning
Proxy feature	A variable that can stand in for another characteristic
Correlation	An observed association, not proof of cause
Sensitive attribute	A characteristic requiring particular care in a context
Ablation	A comparison that removes a feature or component

STUDENT LESSON · 6 OF 18

Follow a worked example

> Follow the worked trace for Proxy Features and Fairness, predict a step, then compare with its explanation.

WORKED EVIDENCE

Reason Step by Step

Predict first. Then check each step. A correct answer without visible **evidence** cannot be audited.

WEEK 32 • WORKED EVIDENCE

1. Photo background predicts classroom location and is correlated with camera quality.
2. Model uses background as a shortcut for health labels.
3. Create paired crops with the same leaf and different backgrounds; predictions change.
4. Crop backgrounds or improve data, retrain, and compare performance and subgroup gaps.

Step 1

STEP 1 Photo background predicts classroom location and is correlated with camera quality.

Step 2

STEP 2 Model uses background as a shortcut for health labels.

Step 3

STEP 3 Create paired crops with the same leaf and different backgrounds; **prediction**s change.

Step 4

STEP 4 Crop backgrounds or improve data, retrain, and compare performance and subgroup gaps.

CHECK Verify the denominator, unit, split, class definition, and data version before interpreting a score.

Recalculate, retrace, or restate the **evidence** independently:

A closer explanation

Proxy features and fairness at a glance

PART	EVIDENCE OR ACTION	INTERPRETATION
North examples	8 X of 10	2 other devices
South examples	2 X of 10	8 other devices
Review question	What does code represent?	Check context and errors

A feature may be legitimate for one purpose and inappropriate for another. Ask what is measured, when it is available, why it predicts and who might be disadvantaged. A correlation alone neither authorizes use nor proves harm. Talk through the intended decision and available alternatives.

Why might removing building but keeping device code leave related information? Explain the steps or evidence behind your response.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Proxy Features and Fairness

Removing an obvious setting field may leave another field that carries the same information.

The facts we will use

Supplied facts

Toy table has location=greenhouse/desk and camera_code=G/D. Every G record is greenhouse; every D record is desk. The team drops location.

Supplied facts

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

AI ACADEMY / AI-4/32

Proxy Features and Fairness

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Remove direct field

The location column is absent from the new input table.



2. Check retained proxy

camera_code still identifies the setting perfectly in this supplied sample.



3. Audit effect

Compare controlled predictions and held-out setting slices; dropping one column does not establish setting independence.

CHANGE ONE THING

Camera codes are assigned independently of setting in a new balanced dataset.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Remove direct field	The location column is absent from the new input table.
2. Check retained proxy	camera_code still identifies the setting perfectly in this supplied sample.
3. Audit effect	Compare controlled predictions and held-out setting slices; dropping one column does not establish setting independence.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

Read the picture in order. Compare controlled **prediction**s and held-out setting slices; dropping one column does not establish setting independence.

The arrows show which recorded **input** or intermediate result each decision uses. Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

What this example does not establish

Keep the stated limit

A correlated field is not necessarily harmful in every use; assess task relevance, consequences and actual model behaviour.

Keep the stated limit

Read the labelled picture: What **evidence** contradicts the original “all setting information removed” claim? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Keep the stated limit

Change one thing: Camera codes are assigned independently of setting in a new balanced dataset. Predict the result before checking the explanation. What remains the same?

STUDENT LESSON · 7 OF 18

Find and repair a mistake

> Find the unsupported step in the Proxy Features and Fairness claim and explain a repair using the supplied evidence.

Find the first error

ERROR DETECTIVE

Compare → Diagnose → Repair

Do not patch the answer. Find the earliest unsupported assumption, data problem, or experiment error.

!

COMPARE • DIAGNOSE • REPAIR

Tempting claim: Removing a sensitive column guarantees the model cannot use related information. Inspect the dataset, split, calculation, error type, subgroup, or experiment record; identify one testable cause; change one factor; rerun the same evidence.

Evidence record	Expected	Actual	Likely cause / next test
normal case	passes		
boundary case	defined behavior		
missing case	safe fallback		
shifted case	measured limit		

CLAIM TO CHALLENGE Removing a sensitive column guarantees the model cannot use related information.

Diagnose one cause and name the cleanest next experiment:

A closer explanation

Do not advertise **fairness** through unawareness merely because names or a sensitive field were removed. A classroom audit should state unresolved questions, intended non-use and a human review path.

Never collect sensitive student information simply to fill this exercise.

> **Does unchanged accuracy after removing a proxy prove fairness?**

STUDENT LESSON · 8 OF 18

Practise together

> Practise Proxy Features and Fairness with support, then explain the deciding step in your own words.

ML STUDIO

Specify Before You Run

A trustworthy result can be reproduced from the recorded data, code, settings, and evaluation rule.



Experiment record

Question to explain

QUESTION + intended use + success threshold

DATA version + grouped split + leakage check

CODE/pre**process**ing + features + model + settings + seed

METRICS + positive class + units + denominators

PREDICTIONS + errors + subgroup slices + limitations

HUMAN reviewer + release, revise, limit, or stop decision

Write the minimum record needed to reproduce today's result:

A closer explanation

Compare versions with and without device code on the same validation cases, while checking subgroup errors and useful task performance.

This **ablation** measures a change in that experiment; correlated **input**s may still carry similar information. Retain group definitions and denominators, and avoid using the final test for repeated feature decisions.

Practise with less help: Find a proxy risk

Use workbook W2 for the connected example "**Proxy feature**s and **fairness**". First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for **Proxy feature**s and **fairness**. Proposed explanation: "A final answer is enough". Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the picture

What **evidence** contradicts the original "all setting information removed" claim? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

> **My explanation using a pictured step and the given facts**

This is supported practice. Keep the separately named independent case for an attempt before feedback.

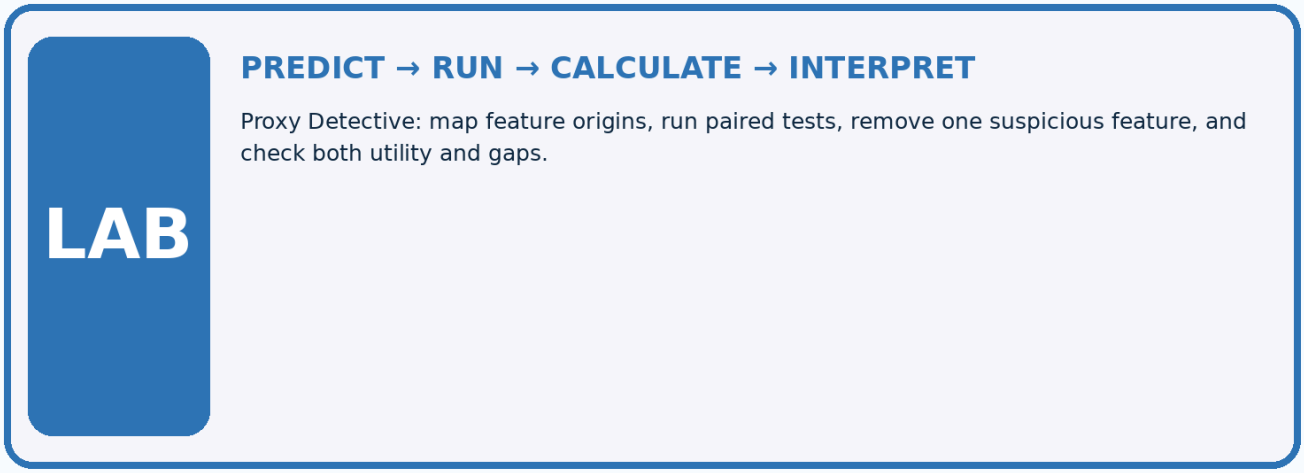
Run the investigation

> Investigate Proxy Features and Fairness. Record the prediction, actual observation or labelled paper trace, explanation and next test separately.

EVIDENCE LAB

Predict → Run → Calculate → Interpret

Keep the test conditions fixed, record every result, and change only one planned factor.



1. Write the hypothesis and expected **evidence** before running.

2. Proxy Detective: map feature origins, run **paired test**s, remove one suspicious feature, and check both utility and gaps.

3. Run one normal, one boundary, one missing, and one shifted case.

4. Calculate the metric from raw counts or row-level errors.

5. Interpret what changed, what did not, and what remains unsafe or unknown.

Case · Prediction · Actual · Calculation / evidence · Meaning / next test

> **Case:** Normal
> **Prediction:** _____
> **Actual:** _____
> **Calculation / evidence:** _____
> **Meaning / next test:** _____

> **Case:** Boundary
> **Prediction:** _____
> **Actual:** _____
> **Calculation / evidence:** _____
> **Meaning / next test:** _____

> **Case:** Missing
> **Prediction:** _____
> **Actual:** _____
> **Calculation / evidence:** _____
> **Meaning / next test:** _____

> **Case:** Shifted
> **Prediction:** _____
> **Actual:** _____

> **Calculation / evidence:** _____
>
> **Meaning / next test:** _____

Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

STUDENT LESSON · 10 OF 18

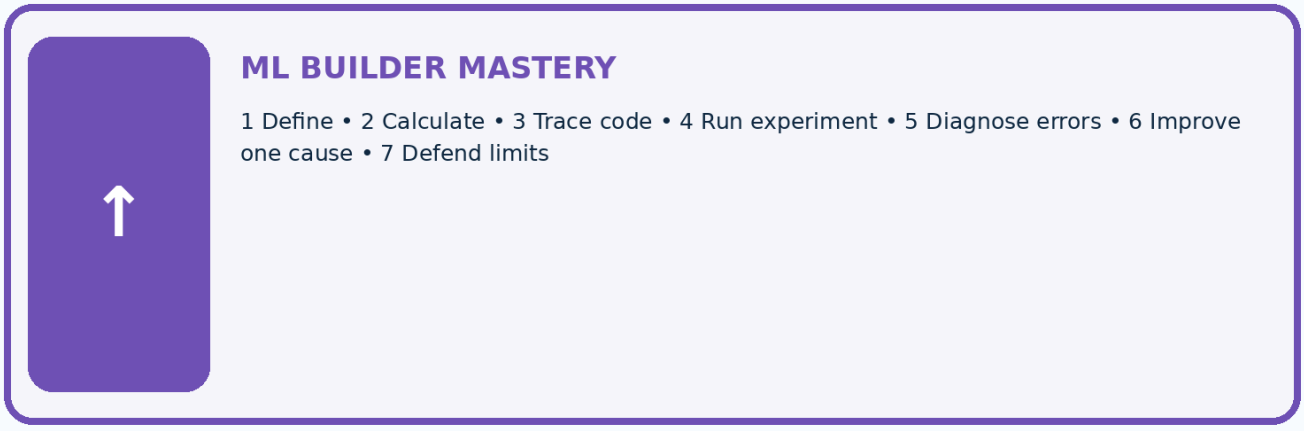
Build your understanding

> Move from recall to tracing, diagnosis and transfer in Proxy Features and Fairness. Explain why each step is justified.

MASTERY

From Knowing the Word to Owning the Idea

Move upward only when you can produce **evidence** without copying the example.



DEFINE Use all four terms precisely: **proxy feature**, **fairness**, **paired test**, **mitigation**.

CALCULATE Reproduce the worked **evidence** with units and denominator.

TRACE Explain the data, code or decision path.

DIAGNOSE Removing a sensitive column guarantees the model cannot use related information.

TRANSFER Solve a fresh dataset or model case.

IMPROVE Change one cause and rerun fixed **evidence**.

DEFEND State intended use, metric, limitation, and human responsibility.

MASTERY RULE A coached calculation becomes mastery only after a different case is completed independently.

STUDENT LESSON · 11 OF 18

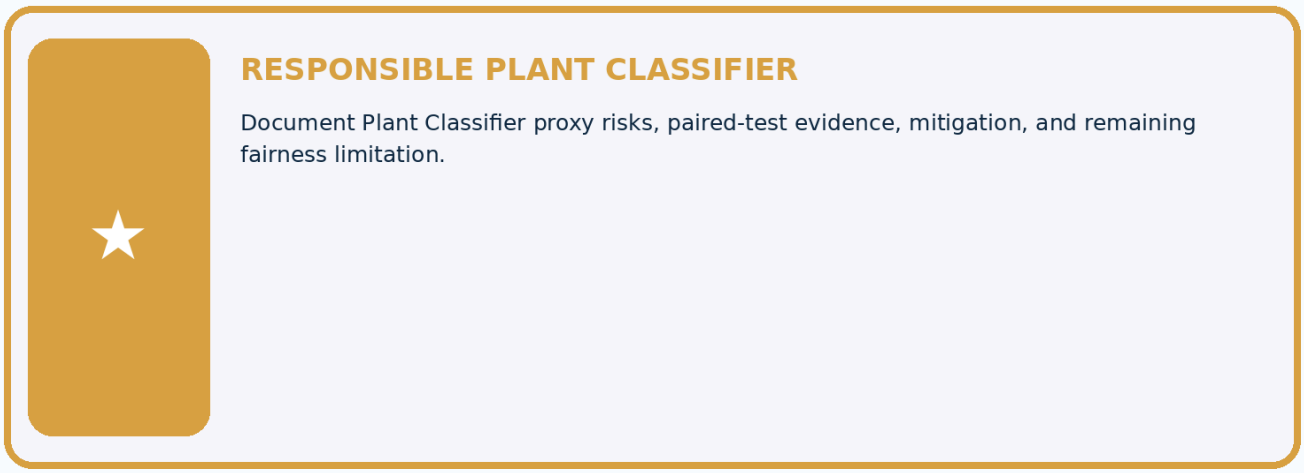
Connect your project

Apply this week's idea to Responsible Plant Classifier. Keep an annotated decision and the evidence supporting it.

CUMULATIVE PROJECT

Responsible Plant Classifier

Every week adds one reviewable artifact to the same **evidence**-based capstone.



THIS WEEK'S CHECKPOINT Document Plant Classifier proxy risks, paired-test **evidence**, **mitigation**, and remaining **fairness** limitation.

Which requirement or risk does this artifact address?

Which dataset, split, code, and metric version does it use?

What normal and shifted cases were tested?

What remains uncertain or weak?

What decision still belongs to a teacher or qualified human?

RESPONSIBLE RELEASE Never use this classroom prototype for medical, pesticide, agricultural, or other consequential decisions.

My **evidence**-linked project decision:

Evidence for your project

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

STUDENT LESSON · 12 OF 18

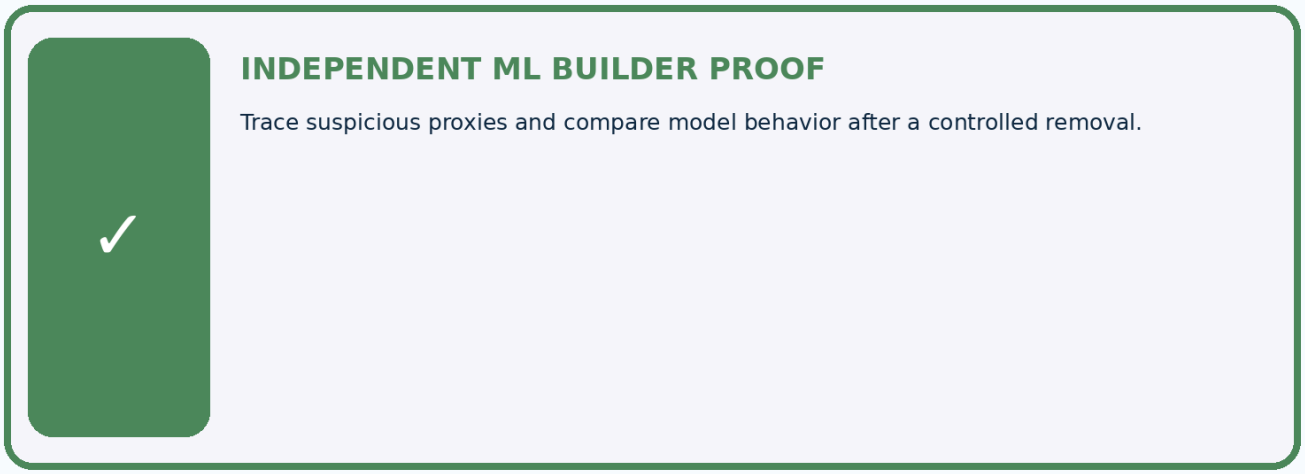
Show what you know

> Use a genuinely new case for independent evidence. Previously discussed examples remain practice; use the separately named independent case before feedback.

INDEPENDENT PROOF

Explain • Calculate • Transfer • Defend

Complete this with a fresh case after guided practice and compare it with the mastery criteria.



What success looks like

REQUIRED PROOF Trace suspicious proxies and compare model behavior after a controlled removal.

- ☐ Define the task, feature/target roles, and intended-use boundary.
- ☐ Show the calculation, code trace, or experiment record.
- ☐ Test a normal case and one revealing boundary, missing, or shifted case.
- ☐ Diagnose one failure without changing several factors at once.
- ☐ Name a limitation, **fairness** concern, and the human-review action.

Independent **evidence** and defense:

My level: ☐ Not yet ☐ With one hint ☐ Independently ☐ I can teach and defend it

Connect to the next lesson

NEXT BRIDGE Carry the experiment **evidence** into: Model Cards and Human Review.

Study a complete case

> Revisit the complete Proxy Features and Fairness case for supported practice. Repeated examples do not become fresh assessments.

Week 32 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

Background colour tracks collection location, associated with plant types.

The question

Explain proxy risk and test.

Worked explanation

Background can be a location shortcut rather than leaf **evidence**. Test leaf types across backgrounds and locations and compare slices.

Explain it in your own words

> Which detail supports the answer, and why does it matter?

Write or draw your reasoning and evidence.

Trace the supplied case visually

Test whether a proxy replaces leaf evidence

Evidence or stage	Value or result
Background colour	Tracks collection location
Location	Associated with plant type
Risk	Model uses shortcut
Test	Cross leaf types with backgrounds and locations

Background colour tracks collection location, associated with plant types.

Explain proxy risk and test.

Background can be a location shortcut rather than leaf **evidence**. Test leaf types across backgrounds and locations and compare slices.

STUDENT LESSON · 14 OF 18

Try a new case independently

> Attempt the reserved new case before feedback. Record support; the earlier examples below remain additional practice.

Independent case: Apply independently

A team removes postcode but retains school code and claims all location-related information is gone. Evaluate the claim and request **evidence**.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Week 32 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

> A plant classifier uses camera_model, and one species was photographed mostly on a single camera. What shortcut might it learn and how could you investigate?

My answer, with a trace, calculation or supporting **evidence**

Write or draw your reasoning and evidence.

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed **input** and expected result

Write or draw your reasoning and evidence.

Connect to your project

Document Plant Classifier proxy risks, paired-test **evidence**, **mitigation**, and remaining **fairness** limitation.

The **evidence** I will keep

Write or draw your reasoning and evidence.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice **evidence** together.

Transfer practice from the original programme

A leaf model performs well using location ID alone. Explain a possible proxy problem and design a diagnostic evaluation.

Explain the mechanism, evidence and one limit.

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

STUDENT LESSON · 15 OF 18

Your lesson route

> Use this map to navigate Proxy Features and Fairness, connect the goals to tasks and keep one evidence record for each attempt.

Start with this question: Removing an obvious setting field may leave another field that carries the same information.

By the end, I can	How I will show it
Identify possible proxy features	Explain the deciding step, show a trace or test, and state one limit.
Explain why removing one field does not guarantee fairness	Explain the deciding step, show a trace or test, and state one limit.
Propose a contextual review with concrete evidence	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

STUDENT LESSON · 16 OF 18

Finish and return

> Explain Proxy Features and Fairness without notes, identify a remaining question and plan a later check with a changed case.

Close your notes. Explain this goal in your own words: Identify possible **proxy feature**s

> **My deciding step, evidence and one remaining question**

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

STUDENT LESSON · 17 OF 18

Supported practice, repair and portfolio

> Attempt supported questions before using hints or feedback. Review a missing building block, then keep the completed work.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Removing an obvious setting field may leave another field that carries the same information.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Compare controlled **prediction**s and held-out setting slices; dropping one column does not establish setting independence.

1. Remove direct field

2. Audit effect

3. Check retained proxy

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Audit effect Compare controlled **prediction**s and held-out setting slices; dropping one column does not establish setting independence. Read the picture in order.

Compare controlled **prediction**s and held-out setting slices; dropping one column does not establish setting independence. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Camera codes are assigned independently of setting in a new balanced dataset. Which conclusion is supported by the supplied facts?

1. camera_code no longer perfectly identifies setting in that dataset.

2. We can decide the result without examining the changed **input** or the procedure.

3. This one example guarantees correct results for every future **input**.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: camera_code no longer perfectly identifies setting in that dataset. The observed mapping changed; other retained fields may still contain setting information.

Repair a missing building block

Review Subgroup Performance (ai-4/31). Goal: Calculate subgroup results with sample sizes. Short practice: Why should the report retain subgroup rates alongside the total?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Feature Engineering (ai-4/29). Goal: Create features available at **prediction** time. Short practice: What should be saved so another group transforms the query identically?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Reproducible Experiments (ai-4/26). Goal: Record enough detail to reproduce a run. Short practice: What should accompany the mean **70%**?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Keep your completed work

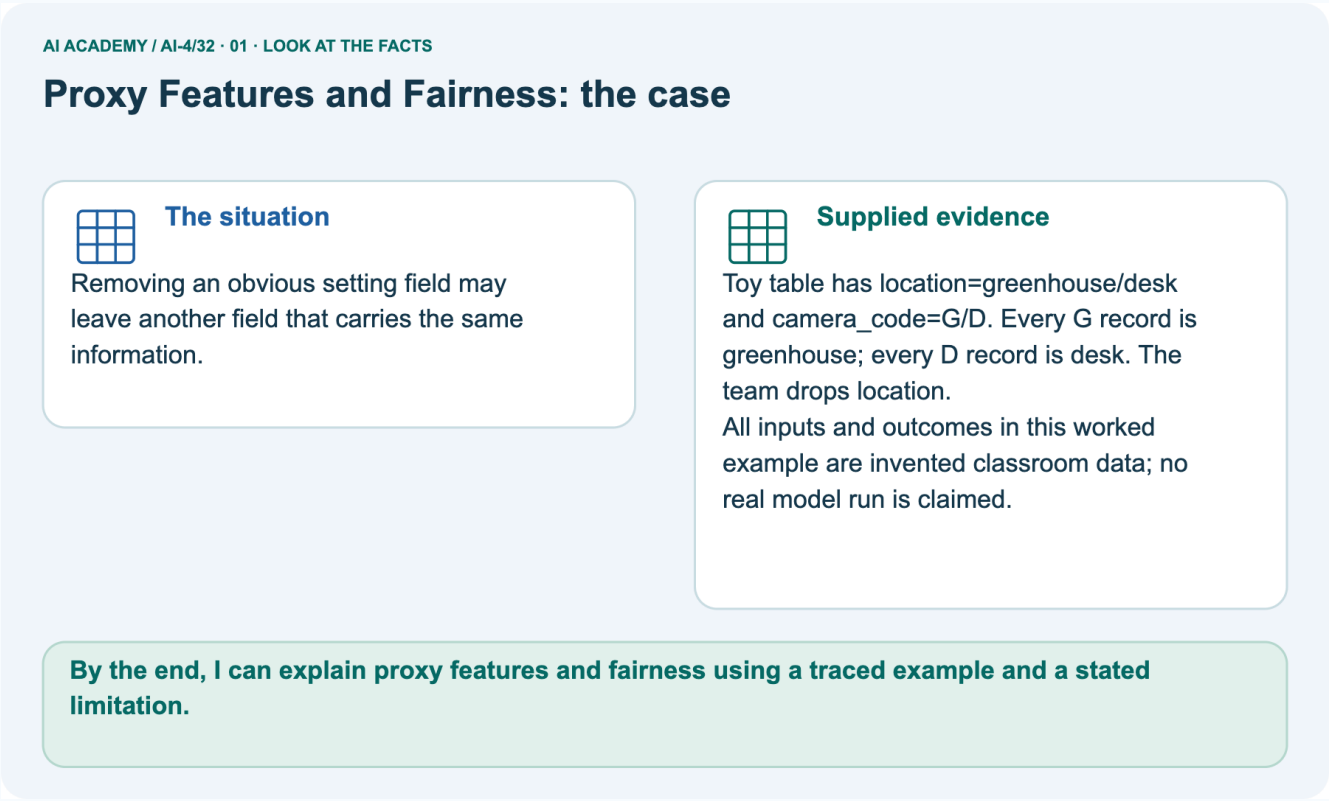
Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

Visual concept journey

> Explain each Proxy Features and Fairness graphic using its labels, deciding evidence and relevant limitation.

01. Proxy Features and Fairness: the case



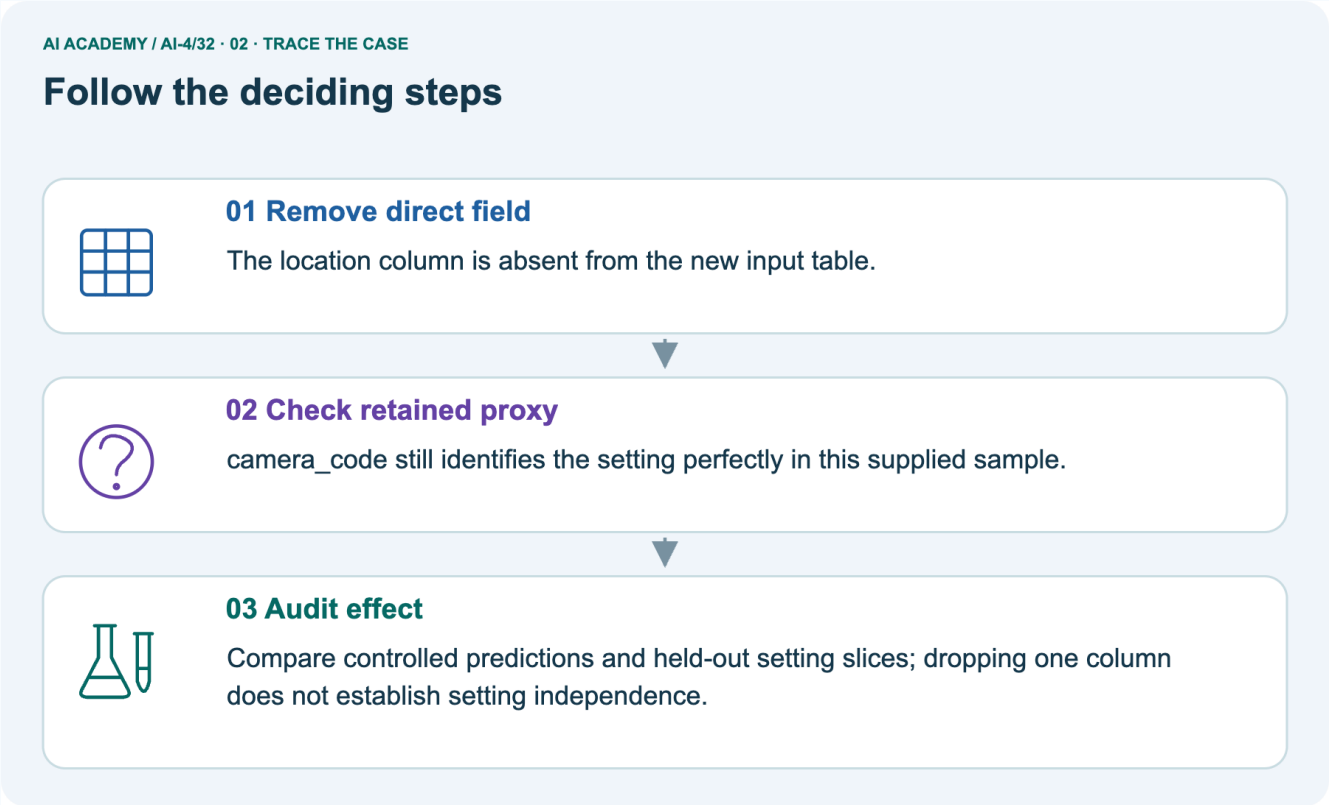
Follow the labelled picture with the matching explanation. Open the vector diagram

Removing an obvious setting field may leave another field that carries the same information. Supplied facts: Toy table has location=greenhouse/desk and camera_code=G/D. Every G record is greenhouse; every D record is desk.

The team drops location. All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

> Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

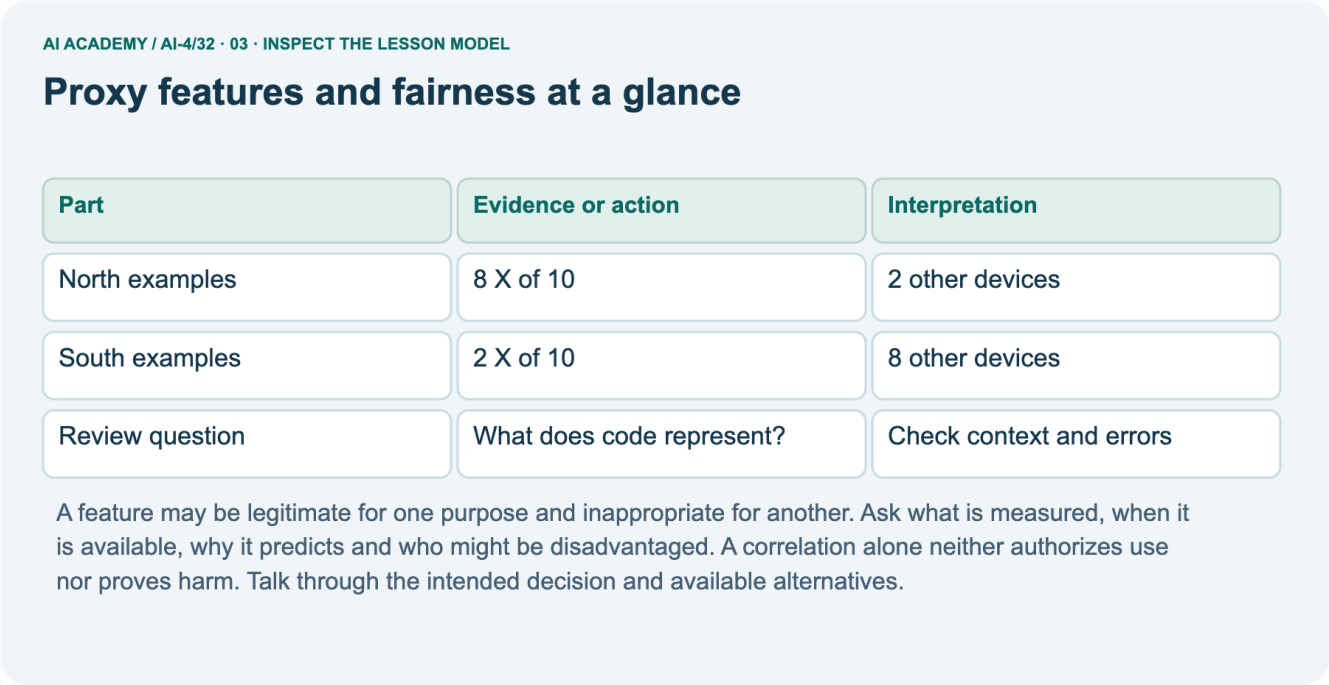


Follow the labelled picture with the matching explanation. Open the vector diagram

Remove direct field: The location column is absent from the new **input** table. → Check retained proxy: camera_code still identifies the setting perfectly in this supplied sample. → Audit effect: Compare controlled **prediction**s and held-out setting slices; dropping one column does not establish setting independence.

> Point to each numbered stage and explain which recorded **input** it uses.

03. Proxy features and fairness at a glance

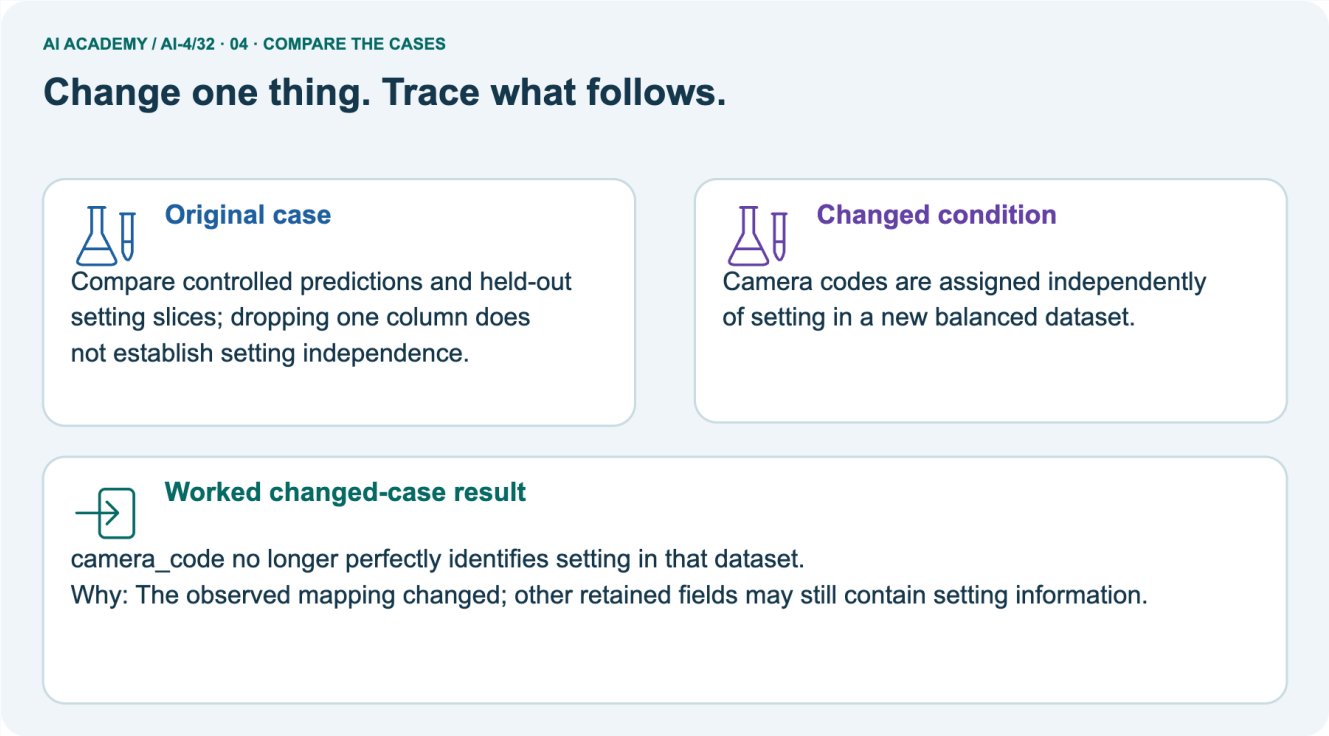


Follow the labelled picture with the matching explanation. Open the vector diagram

Proxy features and **fairness** at a glance. North examples | 8 X of 10 | 2 other devices. South examples | 2 X of 10 | 8 other devices. Review question | What does code represent? | Check context and errors

> Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

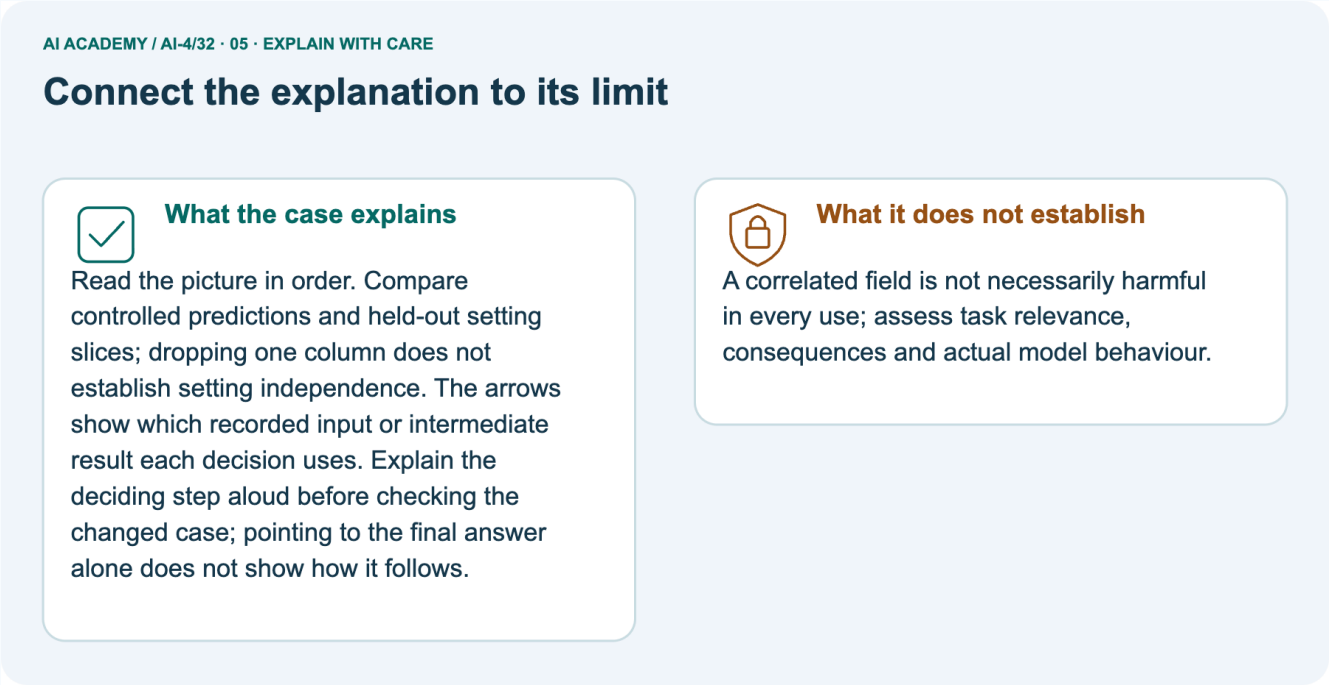
Original case: Compare controlled **prediction**s and held-out setting slices; dropping one column does not establish setting independence.

Changed condition: Camera codes are assigned independently of setting in a new balanced dataset. Expected result: camera_code no longer perfectly identifies setting in that dataset.

Reason: The observed mapping changed; other retained fields may still contain setting information.

> Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

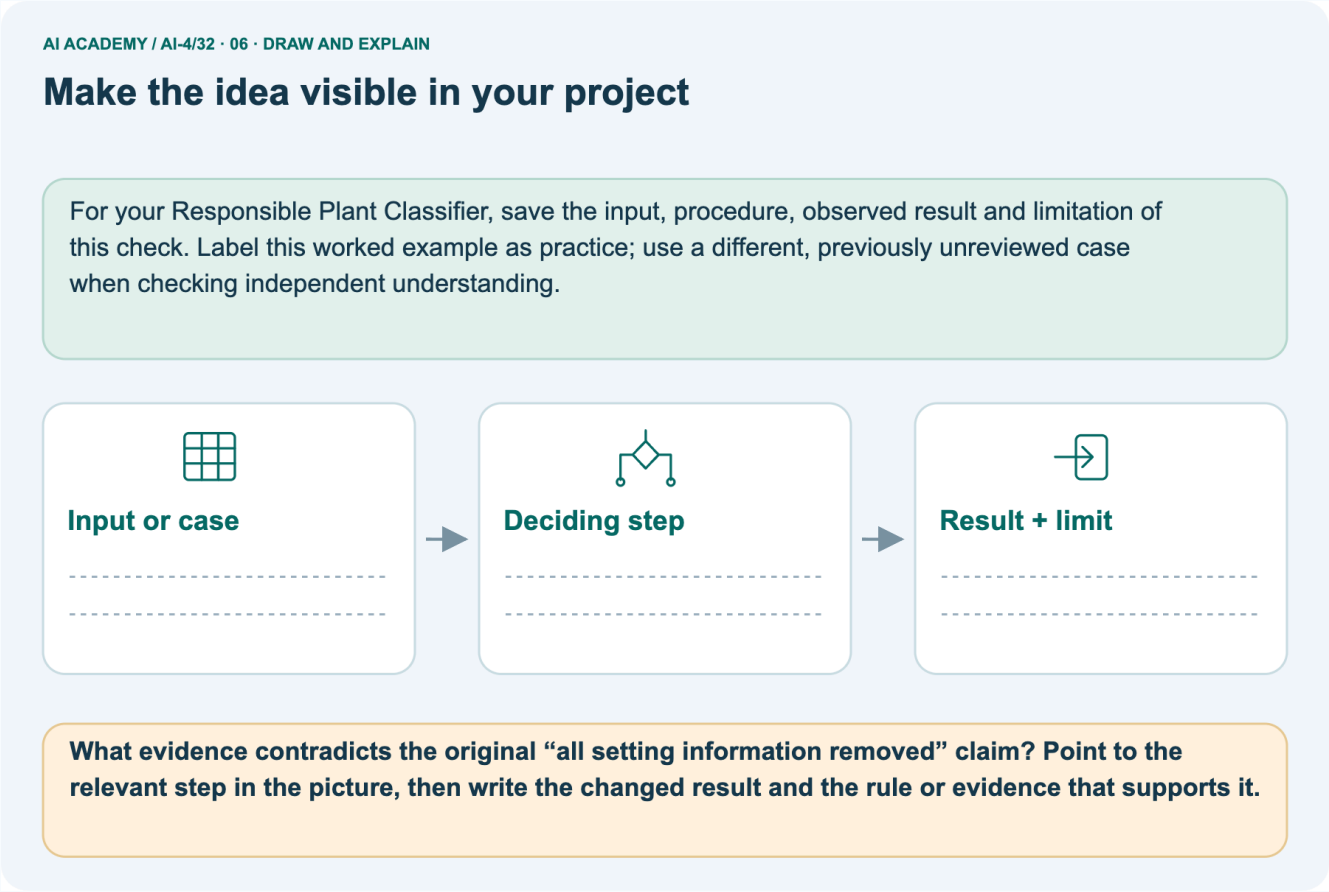
Read the picture in order. Compare controlled **prediction**s and held-out setting slices; dropping one column does not establish setting independence.

The arrows show which recorded **input** or intermediate result each decision uses. Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Limit: A correlated field is not necessarily harmful in every use; assess task relevance, consequences and actual model behaviour.

> Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: What **evidence** contradicts the original “all setting information removed” claim? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

> Draw your project trace and justify the deciding step using **evidence**.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.